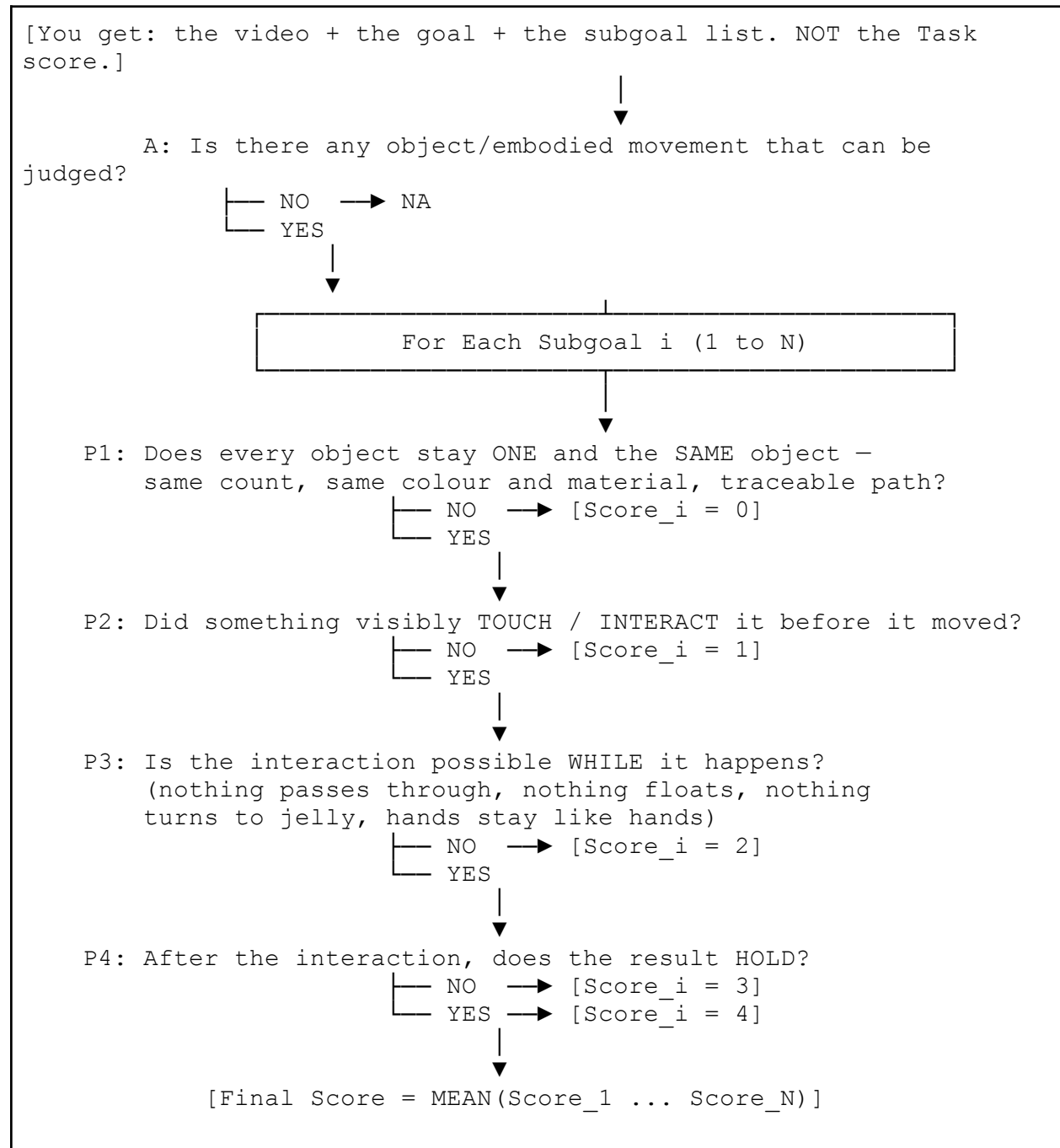


Physical Aspects Human Rubric

Input → Video + Goal | Output → Float[0-4]



Focus strictly on whether what you see could happen in the real world. Reuse the subgoal you break down in the task rubric and then use them **only to know which objects to watch**. Whether

the task actually succeeded is out of scope. A video can complete every subgoal perfectly and still score 0 here. A video can fail every subgoal and still score 3 here.

Step 1: Find Each Subgoal's Window (That Your defined During Task Scoring)

Step 2: Is There Any Object / Embodied Movement That Can be Judged?

1. From the subgoals / window we have mentally tracked from Step 1, is there at least one subgoal where you can see movement of any objects from cause, through movement, to result — for long enough to inspect?
2. Core heuristic
 - a. **Pass (PROCEED TO SCORING):** You can see the hand, the object, and the destination before, during, and after. Count as a yes even if the object is flying or doing weird stuff, it will be checked at the latter filter.
 - b. **Fail (N/A):** The clip is basically static, none of the subgoals' objects are ever handled, or every window is cropped, blocked, or motion-blurred from start to finish.

A video where nothing was attempted is NA, never 3. Physical coherence over an empty scene is not a finding. Do not reward a video for avoiding the task.

NA is not the same as 0. The Task rubric gives 0 when nothing was attempted. Physics gives NA for that, and saves 0 for a video that is maximally broken — which takes a lot of things happening badly. A static clip is Task 0 and Physics NA.

Step 3: The Four Gates

For each subgoal with a window, check **P1 → P2 → P3 → P4 in order**. The moment a gate is Violated, then stop.

Why the order matters: it follows the causal story. If the object was never really one / consistent enough object (P1), asking whether it was pushed properly (P2) is meaningless. If nothing pushed it (P2), asking whether the push looked right (P3) is meaningless. If there is no push / interaction, asking the final state precision (P4) is also not meaningful.

Two possible answers per gate: Yes or No

P1: Does Each Object Stay the Same Object?

1. First check, **if this is 2nd subgoal forward**, w.r.t.t. The most recent previous subgoal's end state, are all objects consistent?
2. Then through this event window, does every object in the scene also stay **one single object**, in a **stable count**, with the **same colour, texture and material**, moving along a **path you can trace**?
3. Core heuristics
 - a. **Pass (Supported)**: One container, present the whole time, same colour, and it only changes position by travelling through the space in between.
 - b. **Fail (Violated)**: It vanishes and comes back. A second copy appears. It turns into a different object. It arrives somewhere without crossing the space in between. It changes colour or material with nothing touching it. It is inconsistent with the previous subgoal's object attribute / definition.

What to look for:

What You See	Plain Name	Code
At least one object changes invalidly w.r.t. Previous subgoal scene (i.e. the bottle shape changes, the napkin changes color)	World inconsistency from the most recent previous subgoal	W1
Object vanishes into nothing	Disappearance / Flickering*	C1
Object pops back into existence	Reappearance / Flickering*	C1
A second copy appears	Duplication	C1
It becomes a different object	Identity swap	C1
Object count changes with no cause	Unlicensed count change	C1
Object grows or shrinks with no cause	Considerable size change	C1
Colour, texture or material changes with nothing touching it	Considerable attribute change	C2
It jumps from A to B without crossing the middle	Teleport / path break, NOT FLYING, BUT "BLIP" LIKE TELEPORT	C1

The Flicker Rule*

The Instability	What To Do
Shorter than about half a second , or the object flickers in appearance but stays in place	Not a physics violation.
Longer than about half a second , or the object is gone long enough that it should have been somewhere	C1 (disappearance / reappearance). Score it as a physics violation.

ONLY USE BELOW SECTION IF YOU ARE ABOUT TO FLAG A COUNT OR APPEARANCE CHANGE

Many correct videos legitimately change how many objects exist. Cutting one thing into two is not a mass violation. Before flagging anything in the count/appearance family, **check the list below**.

- Licensed changes — **these are Supported, not violations**:
 - Splitting** — cutting, tearing, breaking, snapping one thing into several. (cutting a chocolate bar, cutting paper, cracking an egg)
 - Joining** — stapling, assembling, nesting several things into one. (stapling paper, nesting matryoshka dolls, capping a bottle)
 - Using up** — matter legitimately destroyed or absorbed. (erasing a whiteboard, wiping a stain, squeezing out toothpaste)
 - Adding** — matter legitimately added from a visible source. (pouring milk, spraying oil, spreading cream)
 - Revealing** — a lid or cover is removed and exposes things that were already implied to be inside. (opening a shoe box, unzipping a pouch)
- It is a violation when the process is missing, when the result does not match the process shown, or when used-up matter comes back.
- The subgoal's verb licenses its own process and nothing more**. A subgoal "cut the chocolate bar" excuses the bar splitting **once, inside its window, under a visible knife**. It does not excuse the bar splitting again later, and it does not excuse anything else in the frame dividing.

Visual Scene	What Happens	P1 Result
Clean Cut	Knife cuts down through the chocolate bar; one bar becomes	Supported (licensed splitting)

	two pieces.	
Cut With No Knife	The bar separates into two pieces with nothing touching it.	Violated (unlicensed count change)
Wrong Arithmetic	One single downward cut, but the bar ends up in four separate pieces.	Violated (result doesn't match the process)
Second Free Split	The bar is cut once, then a piece splits again on its own later in the clip.	Violated (the verb licensed one cut, not two)
Erased Ink Returns	Marker line is wiped away, then the same line is back on the board a second later.	Violated (used-up matter returned)
Box Opened	Shoe box lid comes off and two shoes are now visible inside.	Supported (revealing)
Spawned Bottle	A second identical bottle appears on the desk next to the first.	Violated (duplication)
Colour Shift	The mug sitting untouched on the desk changes from white to blue.	Violated (attribute change)
Hidden Behind Hand	Bottle is not visible for a moment because the hand covers it, then is visible again.	Supported (normal occlusion — NOT a violation)

P2: Does the motion be executed by any agent / interaction by another object?

1. Look at the exact frames where the object **starts** to move. Does something visibly touch / trigger the motion **before** it responds? **You do not need to care whether the interaction is completely accurate physically, P2 is to measure whether there is a valid trigger that begins the motion, P3 will handle the physics accuracy.**
2. TL;DR → P2 measure action, P3 will later measure reaction accuracy. So even if the aftermath reaction does not make sense, P2 is a Pass.
3. Core heuristics
 - a. **Pass (Supported):** The fingers close on the container, and the container begins to rise only after they make contact. The domino bumps into another domino.
 - b. **Fail (Violated):** It starts moving before anything touches it. It slides with nothing on it. It rises with nothing under it. The hand's grip closes on empty air but the object moves anyway.

What to look for:

What You See	Plain name	Code
Object moves with nothing touching it	Uncaused motion	C3
The thing touching it isn't where the force should be	Contact mismatch	C3
Object starts moving before contact lands	Premature response	C3
Grip closes on empty space but the object still moves	Absent cause	C3

Visual Scene	What Happens	P2 Result
Normal Grasp	Fingers contact the bottle at 0:02.3; the bottle begins rising at 0:02.4.	Supported
Early Lift	The bottle begins rising at 0:02.1; the fingers reach it at 0:02.4.	Violated (premature response)
Ghost Grip	The hand closes 3 cm to the left of the bottle, but the bottle lifts and follows.	Violated (absent cause)

Self-Opening Drawer	The drawer slides open before the hand arrives at the handle.	Violated (uncaused motion)
Blurred Contact	Heavy motion blur over the exact contact frames; you cannot tell if contact happened first.	Unobservable (NOT a violation)

P3: Is the Interaction Itself Possible?

This is the biggest gate and where most real violations live. Watch the **middle** of the action, not the start and end.

1. While the hand and the object are coupled together, does everything behave the way real matter, real mechanisms, and real hands behave?
2. Core heuristics
 - a. **Pass (Supported):** See failure table, if there is nothing.
 - b. **Fail (Violated):** Anything from the list below.

Judge the transition, not the endpoints. A believable first frame and a believable last frame tell you nothing about the frames in between. This is where you should be stepping through frame by frame.

What to look for – go through all of these:

What You See	Plain name	Code
A hand, tool, or object passes through solid matter	Passed through solid	C5
Something gets into or out of a closed container with no opening	Boundary violation	C5
An object rests or moves with nothing holding it up	Floating / no support	C4
A held object stops moving with the hand, or keeps moving with it after release	Decoupled motion	C3
Two things make contact and neither one reacts	No response to contact	C3
Speed or acceleration no visible force could produce; stops dead with no reason	Impossible dynamics	C3
A rigid thing bends like rubber; jelly-like wobble; volume changes	Impossible deformation	C6
Liquid flows upward, clumps, vanishes, or appears with no source	Impossible liquid	C7
A hinge, lid, drawer, or zipper moves along an axis it cannot move along	Mechanism violation	C8

A jointed part comes off its assembly or fails to drive the part it should	Mechanism decoupling	C8
Extra or missing fingers, impossible joint angles, hands merging or morphing	Impossible hand or body	C9
Something melts, boils, freezes or cooks with no cause — or fails to when it clearly should	Impossible heat/state	C10

ONLY USE BELOW SECTION IF YOU ARE ABOUT TO FLAG SOMETHING PASSING THROUGH SOMETHING ELSE

Going behind is not going through. An object that passes behind another object and comes out the other side is completely normal. This is **the most common false alarm** on this gate.

Before flagging "passed through", ask:

1. Could the object simply be **behind** the other one from this camera angle? → **Supported**
2. Can you see a frame where their outlines genuinely **overlap** — where the object is visibly inside the solid, not just hidden by it? → **Violated**

Visual Scene	What Happens	P3 Result
Behind the Box	The bottle moves behind a cereal box and reappears on the other side.	Supported (occlusion)
Inside the Wall	The container's body visibly overlaps the closed pouch wall for 4 frames.	Violated (passed through solid)
Sealed Entry	The container is outside the pouch in one frame and fully inside the next, with the opening still pinched shut.	Violated (boundary violation)
Rubber Laptop	The laptop lid bends and ripples like cloth as it is opened.	Violated (impossible deformation)
Sideways Lid	The laptop lid rotates around a vertical axis instead of its hinge.	Violated (mechanism violation)
Squashed Cushion	A rigid mug is pressed into a soft cloth; the cloth compresses under it.	Supported (this is correct behaviour)

Six Fingers	The hand shows six fingers while gripping the bottle.	Violated (impossible hand)
Uphill Milk	Poured milk arcs upward out of the jug before landing.	Violated (impossible liquid)
Dead Cup	The hand knocks into a cup with clear force and the cup does not move at all.	Violated (no response to contact)
Floating Bottle	The bottle travels 2 cm from the hand with a visible gap, nothing connecting them.	Violated (floating / no support)
Wrong Bag, Clean Physics	The hand carries the bottle smoothly into the wrong bag. Every contact is believable.	Supported (task problem, not physics)
Stiff Motion	The hand moves in a jerky, robotic way, but every position is physically reachable.	Supported (ugly, not impossible)

P4: Does the Result Hold After the Interaction?

1. Once the hand withdraws or the action ends, does the state it produced stay put the way the visible forces allow?
2. **Judge the state the video actually produced, not the one the subgoal asked for.** If the container was dropped beside the pouch instead of inside it, P4 asks whether it stayed beside the pouch.
3. Core heuristics
 - a. **Pass (Supported):** The container stays where it was released after action ends, and stays there when its support is moved.
 - b. **Fail (Violated):** It falls with nothing acting on it. It rests on nothing. It sinks into a solid surface. A closed lid springs open untouched. The whole thing reverses within a few frames.

What to look for:

What You See	Plain name	Code
The result undoes itself right after release	Outcome reversal	C4
The object rests on nothing, or sinks into a surface	Unsupported rest	C4
A closed lid, drawer, or zip reopens untouched	Closure reopening	C8
The object keeps following the hand after release	Post-release coupling	C3
A mechanism reverses on its own	Mechanism reversal	C8

Visual Scene	What Happens	P4 Result
Stays Put	The container remains in the pouch as the hand withdraws and the pouch is lifted.	Supported
Stays Put, Wrong Place	The container is dropped on the desk instead of in the pouch, and stays on the desk.	Supported (it held — task problem, not physics)
Hovering	The hand withdraws and the container stays floating in mid-air.	Violated (unsupported rest)

Sinking	The mug is set down and slowly sinks halfway into the table top.	Violated (unsupported rest)
Drawer Rebound	The drawer is pushed fully closed, then slides open again with nothing touching it.	Violated (closure reopening)
Sticky Hand	The hand opens and moves away, but the bottle keeps travelling with the palm.	Violated (post-release coupling)